

CAPM analysis Report

1. Introduction of CAPM

This project estimates the Capital Asset Pricing Model (CAPM) for 10 U.S. stocks using daily returns from 2 January 2015 to 31 December 2025. The selected stocks are Microsoft, Alphabet, Amazon, JPMorgan Chase, Johnson & Johnson, Procter & Gamble, Exxon Mobil, Caterpillar, NextEra Energy, and Air Products and Chemicals, covering a range of industries including software and cloud services, healthcare, banking, and energy.

S&P500 is used as the market benchmark. The analysis compares full-sample alpha, beta, explanatory power, and statistical significance, and examines beta stability using 252-day and 126-day rolling windows.

The main analysis is presented in CAPM_analysis.ipynb. It includes a regression table, a beta comparison bar chart, scatter plots with regression lines, residual plots, and rolling-beta charts.

2. Formulas

Returns are defined as simple adjusted-price returns, expressed in decimal form: $R_t = P_t / P_{(t-1)} - 1$. Subtracting the risk-free rate from the return gives the excess return used in the subsequent CAPM analysis.

In addition, the risk-free rate is converted from an annual percentage rate into a daily decimal return using the following formula: $rf["Risk_Free_Rate"] = (1 + rf["DTB3"] / 100)^{**} (1 / 252) - 1$.

In the CAPM framework, we estimate an ordinary least squares (OLS) linear regression with an intercept. The regression specification is $R_{stock,t} - R_{f,t} = \alpha + \beta \times (R_{SPY,t} - R_{f,t}) + \epsilon_t$. Alpha represents the daily intercept in decimal-return units and can be multiplied by 100 to express it as a percentage per day. Beta measures the sensitivity of a stock's excess returns to market excess returns and should not be interpreted as a measure of total stock volatility.

Beta can also be estimated as $\beta_t = \text{Cov}(\text{stock excess return}, \text{market excess return}) / \text{Var}(\text{market excess return})$. This covariance-variance formulation is used to calculate the rolling-beta estimates.

3. Data used and data exploratory

The S&P 500 is used as the market benchmark. From the S&P 500 universe, we select 10 stocks from different industries.

Finally, the 3-month Treasury bill rate is used as the risk-free rate.

Regarding the source for our data, we employ SPY ETF as the proxy for the S&P 500, Tiingo as the source obtaining historical stock prices, and FRED DTB3 3-month Treasury bill secondary market rate as the risk-free rate.

4. Methodology of Research

The Capital Asset Pricing Model (CAPM) is used to analyse the 10 selected stocks and compare their full-sample alpha, beta, explanatory power, and statistical significance. We then examine the stability of beta using 252-day and 126-day rolling windows.

In the model, we first calculate the excess stock return and excess market return through the formula:

Excess stock return = stock return – risk-free rate

Excess market return = S&P 500 return – risk-free rate

Returns and the risk-free rate are expressed on a daily basis. Since the risk-free rate is reported as an annualised rate, it is first converted to a daily rate.

The linear regression specification is $y = \alpha + \beta \times x + \epsilon$, where alpha is the intercept, beta is the slope coefficient, and epsilon represents the error term.

Then, we could analysis the according alpha, beta, R-squared, and p-values. Alpha is daily intercept in decimal-return units; multiply by 100 for percent per day. It is not annualized in the results table. Beta is the sensitivity of stock excess returns to market excess returns; it is not a measure of total stock volatility. R-squared is the fraction of in-sample stock excess-return variation explained by the market factor. Finally, the p-values is the main regression uses ``fit(cov_type="HC3")``, providing heteroskedasticity-robust standard errors. HC3 does not adjust for serial correlation.

We then present a bar chart comparing beta estimates across the selected stocks. This provides a direct comparison of each stock's estimated market sensitivity with the benchmark value of 1.

We also present scatter plots of stock excess returns against market excess returns, together with the corresponding CAPM regression line. Large residuals indicate

observations for which realised excess returns differ substantially from the model's fitted values. A closer concentration of observations around the regression line indicates a stronger linear relationship with the market factor.

To further assess the distribution of residuals, we present residual-versus-fitted-value scatter plots and histograms for each stock. These plots help identify outliers, extreme observations, and departures from a distribution centred around zero.

Finally, rolling-beta charts are used to examine how estimated market sensitivity changes over time. A 252-day rolling window corresponds approximately to one year of trading days. For example, a rolling beta of 1.2 indicates an estimated sensitivity of 1.2 to market excess returns over the corresponding window. The first estimate uses days 1–252, the second uses days 2–253, and so on. This approach allows us to assess whether market exposure remains relatively stable and to identify periods of substantial change that may reflect shifts in market conditions or the firm's exposure to systematic risk.

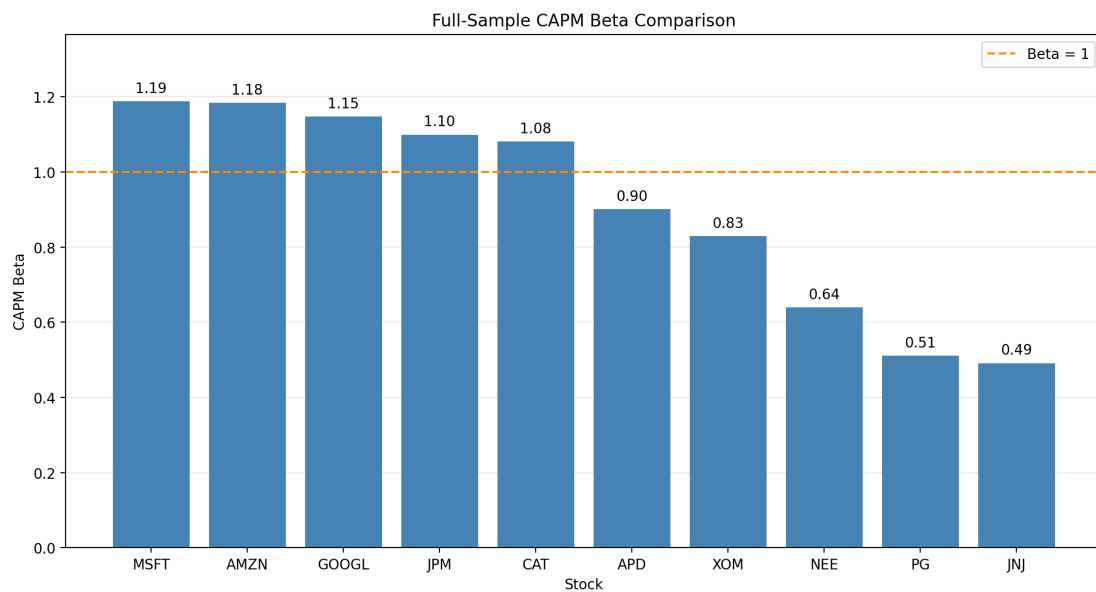
5. Conclusions of results (graphs, numbers, figures, statistics, The R-squared value)

The estimated regression parameters are presented in the chart below. XOM generated lower excess returns than those predicted by the CAPM model, conditional on its estimated market exposure.

R-squared measures the proportion of variation in excess returns explained by the market factor and therefore provides an indication of the model's explanatory power. For example, the market explains a relatively smaller proportion of the variation in JNJ, PG, XOM, and NEE, whose R-squared values are around 0.25. An R-squared of 0.25 indicates that approximately 25% of the variation in excess returns is explained by the market factor in the sample. Two stocks with similar beta estimates can have different R-squared values because stock returns are also affected by firm-specific factors and other sources of variation.

For the alpha p-value, MSFT provides an illustrative example. Its p-value of 0.0487 is below 0.05, so the estimated alpha is statistically significant at the 5% level. In general, a p-value below 0.05 provides evidence against the null hypothesis that the coefficient is zero at the 5% significance level. Conversely, a p-value above 0.05 does not provide sufficient statistical evidence to conclude that the coefficient differs from zero. The beta p-values are approximately zero for all stocks, indicating that the estimated beta coefficients are statistically significant.

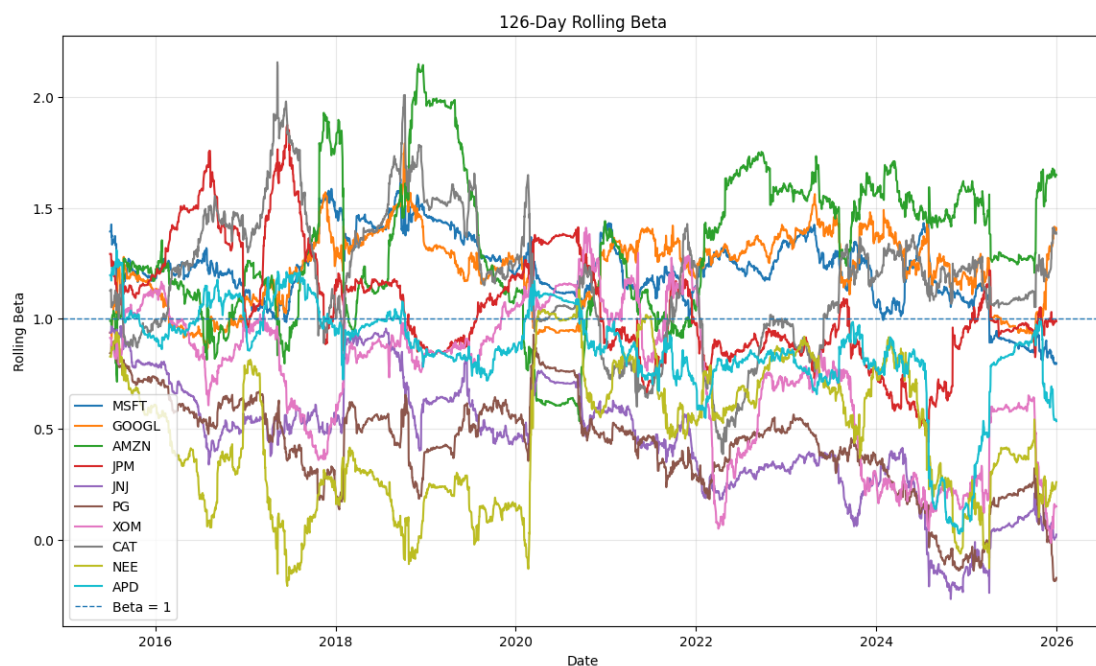
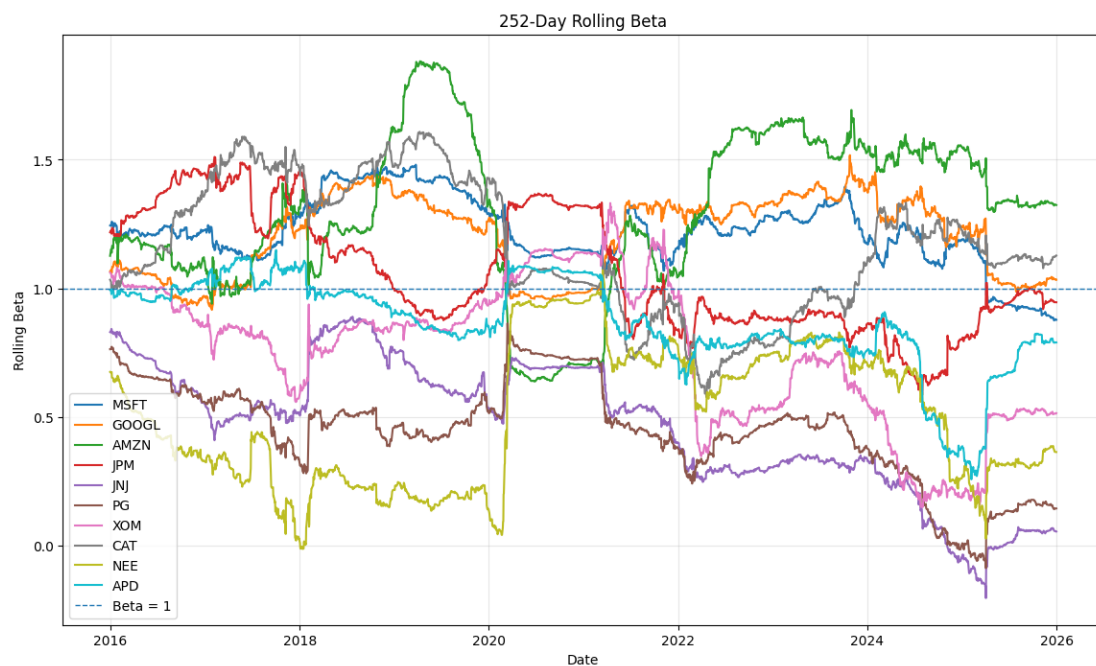
	Stock	Alpha	Beta	R_squared	Alpha_pvalue	Beta_pvalue
1	MSFT	0.000393	1.187671	0.617735	0.048697	0.000000e+00
2	GOOGL	0.000424	1.147625	0.501325	0.082651	0.000000e+00
3	AMZN	0.000537	1.183813	0.409175	0.079112	4.785782e-114
4	JPM	0.000234	1.099207	0.519174	0.297922	2.732633e-181
5	JNJ	0.000107	0.491280	0.230687	0.578091	2.814489e-59
6	PG	0.000017	0.511219	0.241408	0.929628	2.118186e-47
7	XOM	-0.000065	0.828602	0.289202	0.813749	1.888337e-90
8	CAT	0.000340	1.081016	0.412020	0.220002	1.400590e-190
9	NEE	0.000235	0.640170	0.213160	0.371537	2.383775e-44
10	APD	-0.000097	0.900577	0.394200	0.684675	1.076252e-210



The distribution of beta estimates is shown in the diagram. Most beta estimates lie between 0.8 and 1.2, while PG has the lowest beta, at approximately 0.5. This indicates that PG has relatively low sensitivity to market excess returns: a 1% increase in market excess returns is associated with an estimated 0.5% increase in PG's excess returns.

The rolling-beta diagrams for the 252-day and 126-day windows show that the 126-day estimates are more volatile than the 252-day estimates. The blue dashed line represents a beta of 1, corresponding to the market benchmark. A beta greater than 1 indicates greater sensitivity to market movements, whereas a beta between 0 and 1 indicates lower sensitivity. A beta below 0 indicates movement in the opposite direction to the market over the corresponding window. Overall, beta varies over time for all

stocks. NEE exhibits the largest variation, while APD and JPM show comparatively smaller variation in rolling beta.



First observation is that beta is time-varying, namely, the rolling beta estimates show substantial variation over time across the selected stocks. This suggests that market sensitivity is not constant throughout the sample period.

Second observation would be: window length matters. The choice of rolling window affects the estimated beta. The 126-day rolling beta reacts more quickly to recent

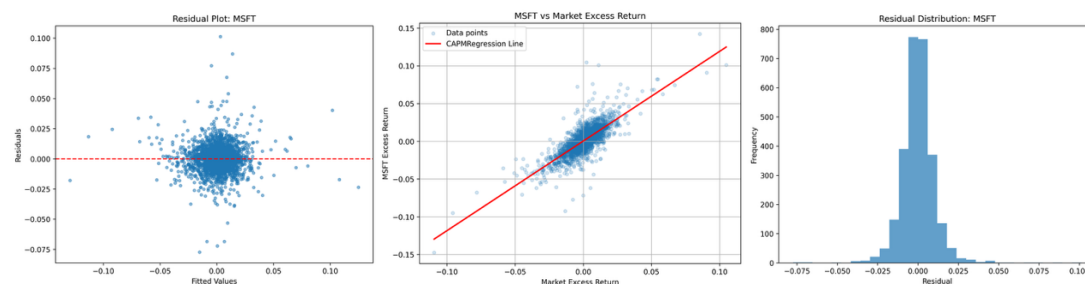
market conditions, while the longer-window beta provides a smoother estimate of systematic risk.

The 252-day rolling-beta window produces a smoother estimate than the 126-day window because it incorporates a larger number of observations and therefore responds more slowly to recent changes. In contrast, the 126-day rolling beta uses a shorter window and is consequently more responsive to recent market conditions, but also more volatile.

6. Short paragraph of findings

Three specific examples are discussed below:

MSFT has an estimated alpha of 0.000393, indicating a positive abnormal return relative to the return predicted by the CAPM model. Its beta of 1.19 indicates relatively high sensitivity to market movements, with a 1% increase in market excess returns associated with approximately a 1.19% increase in MSFT's excess returns. The R-squared of 0.618 indicates that approximately 61.8% of the variation in MSFT's excess returns is explained by the market factor. The alpha p-value of 0.0487 is below the 5% significance level, so the estimated alpha is statistically significant at the 5% level, although only marginally. The beta coefficient is also highly statistically significant.



PG has a beta of 0.51, indicating relatively low sensitivity to market excess returns. A 1% increase in market excess returns is associated with approximately a 0.51% increase in PG's excess returns. However, its R-squared is only 0.241, suggesting that the market factor explains approximately 24.1% of the variation in its excess returns. Its estimated alpha is close to zero, and the alpha p-value of 0.930 is substantially above 0.05, indicating that the alpha is not statistically significant. Therefore, there is insufficient statistical evidence to conclude that PG generated abnormal returns after controlling for market exposure.



XOM has a negative estimated alpha of -0.000065 , suggesting that its excess return was below the level predicted by the CAPM model over the sample period. However, the alpha p-value of 0.814 is far above the 5% significance level, so there is insufficient statistical evidence to conclude that the true alpha differs from zero. Its beta of 0.83 indicates moderate sensitivity to market excess returns, while the R-squared of 0.289 suggests that the market factor explains approximately 28.9% of the variation in XOM's excess returns.



The rolling-beta analysis provides further evidence that market sensitivity varies over time. AMZN exhibits a relatively high and persistent rolling beta, particularly during 2018–2020 and 2022–2025, indicating relatively high sensitivity to market movements. MSFT's market sensitivity also varies over time, with rolling beta generally remaining above 1 while changing substantially across different periods.

In contrast, JNJ, PG, and NEE generally exhibit rolling betas below 1, indicating relatively lower sensitivity to market movements than higher-beta stocks such as AMZN. A single CAPM beta may therefore obscure changes in a firm's market exposure over time. AMZN, MSFT, GOOGL, and CAT have mostly recorded betas above 1, indicating relatively greater sensitivity to market movements. Conversely, JNJ, PG, and NEE have betas below 1, indicating lower sensitivity. JPM also shows a noticeable decline in beta after 2022, followed by a partial recovery towards the end of the sample.

In conclusion, the CAPM model provides a framework for assessing a stock's sensitivity to market movements through its beta. The regression results provide information on the statistical significance and explanatory power of the estimated coefficients, while the rolling-beta analysis demonstrates that market sensitivity can vary over time.